

Fractional indices



1 Write the following in surd form:

a $x^{\frac{1}{3}}$

b $a^{\frac{3}{2}}$

c $20^{\frac{5}{4}}$

d $y^{-\frac{1}{2}}$

e $3x^{\frac{1}{2}}$

f $b^{2\frac{1}{3}}$

g $y^{1.5}$

h $5^{-\frac{4}{3}}$

2 Write the following as a single power of 2:

a $\sqrt[3]{2}$

b $\sqrt[5]{16}$

c $\sqrt{32}$

d $2\sqrt{2}$

e $8\sqrt[3]{2}$

f $2^a \times \sqrt{2}$

g $\frac{8}{\sqrt{2}}$

h $\frac{1}{\sqrt{8}}$

3 Write the following as a single power of 3:

a $\sqrt{27}$

b $3^x \times \sqrt[4]{3}$

c $9\sqrt{3}$

d $\frac{1}{\sqrt[3]{9^x}}$

4 Complete the following statement:

$$\begin{aligned}\sqrt{m^8} &= (m^8)^{\frac{1}{2}} \\ &= m^{8 \times \frac{1}{2}} \\ &= m^{\frac{1}{2}}\end{aligned}$$

5 Write the following in the form x^k , where k is rational:

a $\sqrt{x}$

b $\sqrt[6]{x}$

c $\frac{1}{\sqrt{x}}$

d $(\sqrt[3]{x})^2$

e $\sqrt[4]{x^3}$

f $(\sqrt[7]{x})^6$

g $\sqrt[4]{x^8}$

h $\frac{1}{\sqrt{x}}$

i $\sqrt{\sqrt{x}}$

j $\sqrt[3]{x^{6a}}$

k $\frac{1}{\sqrt{x^{-3}}}$

l $x\sqrt{x}$

m $\frac{1}{\sqrt[4]{x^5}}$

n $\sqrt[5]{\sqrt{x}}$

o $x^2 \times \sqrt[3]{x}$

6 Consider the expression $64^{\frac{2}{3}}$.

a Complete the statement: $64^{\frac{2}{3}} = (\sqrt[3]{64})^{\frac{1}{2}}$

b Hence, evaluate $64^{\frac{2}{3}}$.

7 Without using a calculator, evaluate the following:



a $1^{\frac{1}{10}}$

b $4^{\frac{3}{2}}$

c $121^{\frac{1}{2}}$

d $\sqrt[3]{27}$

e $125^{\frac{1}{3}}$

f $(-64)^{\frac{1}{3}}$

g $64^{-\frac{1}{6}}$

h $81^{-\frac{3}{4}}$

i $\sqrt[3]{-64}$

j $\left(\frac{9}{100}\right)^{\frac{1}{2}}$

k $\left(64^{\frac{1}{9}}\right)^{\frac{9}{2}}$

l $\left(\frac{8}{125}\right)^{\frac{2}{3}}$

m $(-32)^{\frac{4}{5}}$

n $\frac{64^{\frac{1}{3}}}{64^{\frac{2}{3}}}$

o $\frac{1}{-\sqrt{169}}$

p $\sqrt[3]{\frac{-64}{125}}$

q $\frac{\sqrt[3]{40}}{\sqrt[3]{5}}$

r $36^{\frac{1}{2}} - 32^{\frac{3}{5}}$

s $1000^{\frac{8}{9}} \times 1000^{-\frac{5}{9}}$

t $\sqrt[3]{\sqrt[4]{16} + \sqrt{625}}$

8 Use a calculator to evaluate the following, to two decimal places:

a $10^{\frac{3}{2}}$

b $4^{\frac{5}{3}}$

c $\sqrt[4]{5^3}$

d $(\sqrt[3]{12})^5$

9 Simplify the following expressions, giving your answers in index form. Assume that all variables represent positive numbers.

a $(\sqrt{b})^8$

b $\sqrt{m^6}$

c $\sqrt[4]{a^5}$

d $\frac{1}{\sqrt[5]{a^6}}$

e $\sqrt[3]{x^6}$

f $\sqrt[3]{m^3}$

g $\sqrt{j^2 x^4}$

h $\left(\sqrt[4]{x^5 y^3}\right)^{24}$

i $(4a^8)^{\frac{1}{2}}$

j $(625u^{16}v^{12})^{\frac{1}{2}}$

k $8b^{\frac{3}{4}} \div 2b^{\frac{2}{3}}$

l $y^3 \times \sqrt[3]{y}$

m $\sqrt{(2x+9)^2}$

n $\sqrt[3]{9^3 x^{18} y^{12}}$

o $\left(\frac{x^{15}}{1024}\right)^{\frac{2}{5}}$

p $\sqrt{\frac{36x^{18}}{y^{20}}}$

q $\sqrt{16y^2 + 24y + 9}$

r $\sqrt{x^2 y^2 + 18xy^2 + 81y^2}$

10 Simplify the following expressions, giving your answers in surd form. Assume that all variables represent positive numbers.

a $\frac{\sqrt{x^2 + 5x + 6}}{\sqrt{x + 2}}$

b $\frac{\sqrt[3]{x^2 + 9x + 20}}{\sqrt[3]{x + 4}}$

11 Describe how we could interpret the expression $m^{\frac{q}{r}}$ in terms of powers and roots of m .

12 Determine whether the following statements accurately describe the meaning of the expression $x^{-\frac{y}{z}}$:



- a $x^{-\frac{y}{z}}$ means we are raising x to the power of $\frac{z}{y}$, then taking the reciprocal of the result.
- b $x^{-\frac{y}{z}}$ means we are taking the reciprocal of x , then raising the result to the power of $\frac{y}{z}$.
- c $x^{-\frac{y}{z}}$ means we are taking the reciprocal of x , then raising the result to the power of $\frac{z}{y}$.
- d $x^{-\frac{y}{z}}$ means we are raising x to the power of $\frac{y}{z}$, then taking the reciprocal of the result.

13 Is there a real number that equals $\sqrt[4]{-16}$? Explain your answer.

14 Consider the expression $m^5 \times m\sqrt{m}$.

- a Express it in simplest index form.
- b Express it in surd form.

15 Consider the expression $m^{\frac{1}{4}} = 8^{\frac{1}{2}}$.

- a Complete the following working:

$$\begin{aligned} m^{\frac{1}{4}} &= (m^1)^{\frac{1}{2}} \\ &= \sqrt{1^{\frac{1}{2}}} \end{aligned}$$

- b Hence, or otherwise, solve the equation $m^{\frac{1}{4}} = 8^{\frac{1}{2}}$.

16 Consider the equation $x^{\frac{2}{3}} = 9$.

- a Complete the statement: $(x^{\frac{2}{3}})^1 = x$
- b Hence, solve the equation $x^{\frac{2}{3}} = 9$.

17 Find the missing values in the following equations:

- a $16^{\frac{3}{4}} = 64$
- b $(1)^{\frac{2}{3}} = 25$
- c $625^{\frac{3}{4}} = 125$
- d $(1)^{\frac{4}{3}} = 81$

18 Solve the following equations for x :

- a $x^{\frac{3}{4}} = 125$
- b $(x - 4)^{0.5} = 4$
- c $(2x - 1)^{\frac{2}{3}} = 9$
- d $(4x + 3)^{-1.5} = 8$

19 Solve the following equation for k :

$$\sqrt[k]{y} \times \sqrt[k]{y} \times \sqrt[k]{y} = y^{\frac{1}{2}}$$

- 20** To evaluate $81^{\frac{3}{2}}$ would it be more efficient to use the property $a^{\frac{m}{n}} = (\sqrt[n]{a})^m$, or the property $a^{\frac{m}{n}} = \sqrt[n]{a^m}$. Explain your answer.



Applications

- 21** The surface area A of a cube with a volume V is given by $A = 6V^{\frac{2}{3}}$.

- a** Find the surface area of a cube with a volume of 8 cm^3 .
- b** Find the volume of a cube with a surface area of 216 cm^2 .

- 22** The volume V of a sphere with surface area A is given by:

$$V = \frac{A^{\frac{3}{2}}}{6\sqrt{\pi}}$$

- a** Find the volume of a sphere with a surface area of 50 cm^2 . Round your answer to the nearest cubic centimetre.
- b** Find the surface area of a sphere with a volume of 125 cm^3 . Round your answer to the nearest square centimetre.

- 23** The volume V of a regular tetrahedron with edge length a is given by:

$$V = \frac{a^3}{6\sqrt{2}}$$

- a** Find the volume of a tetrahedron with a side length of 2 cm .
- b** Rearrange the formula to make a the subject.
- c** Hence, find the side length of a tetrahedon with a volume of 72 cm^3 .

- 24** A company's net profit (in dollars) can be modelled by the equation:

$$P = 50n^{\frac{3}{4}} - 2500$$

Where n is the number of units sold.

- a** Find its net profit be in dollars if it manages to sell 256 units.
- b** Find the number of units needed to be sold to make a net profit of \$8300.

25 A company's net profit (in dollars) can be modelled by the equation:

$$P = 180n^{\frac{3}{5}} - 4000$$

Where n is the number of units sold.

- a Find its net profit in dollars if it manages to sell 32 units.
- b Find the number of units needed to be sold to make a net profit of \$45 000. Round to the nearest whole unit.